Semiconductor Devices

Engineering Technology

May, 2026

Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

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Semiconductor Devices (A12001)

Short Title:	Semiconductor Devices		
Faculty	Science and Computing		
Department	Engineering Technology		
Cluster	Electronics		
Author	CLAIRE.FITZPATRICK		
Credits	5	**Level:**	Advanced

Description of Module / Aims

The course aims to advance the students knowledge of the fundamentals of solid state theory in respect of the analysis and design of complex semiconductor based transistor structures. A comprehensive review of the design, operation and manufacture of bipolar, metal-oxide-semiconductor and compound transistor structures is undertaken. The use of semiconductor TCAD tools for complex transistor structures is also concurrently introduced.

Programmes

ENGR-0022	BEng (Hons) in Electronic Engineering (WD_EONIC_B)	4 / 8 / E
	BEng (Hons) in Information Engineering (International) (WD_EEELC_BI)	4 / 8 / M
ENGR-0022	BSc (Hons) in Physics for Modern Technology (WD_KPHTE_B)	4 / 8 / M

Indicative Content

- Review of Semiconductor Theory
- The Bipolar Junction Transistor & Applications
- The Metal-Oxide-Semiconductor Transistor & Applications
- Compound Semiconductor Devices & Applications
- Semiconductor TCAD Tools

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Describe the fundamental concepts of solid state theory as applied in the analysis of carrier transport processes within semiconductor materials and transistor devices.
2. Explain the advanced physical processes employed in the fabrication of semiconductor transistor structures.
3. Analyse the basic design, operation and manufacture of bipolar, metal-oxide-semiconductor and compound transistor structures.
4. Apply the knowledge and comprehension gained in analysing and relating macroscopic semiconductor diode performance parameters to semiconductor solid state theory & associated fabrication processes.
5. Demonstrate the basic use of TCAD software tools and methods as used in the design, analysis and development of complex transistor structures.

Learning and Teaching Methods

- Lectures
- Practical / Project Work

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	60%	1,2,3,4
Continuous Assessment	40%	
Practical	40%	5

Assessment Criteria

70% - 100%: The learner has: attained the module learning outcomes at an excellent level; demonstrated a comprehensive knowledge of the associated subject matter; achieved an excellent level of the skills required for the subject matter; demonstrated the ability to carry out further investigation and problem solving associated with the subject matter.

60% - 69%: The learner has: attained the module learning outcomes at a very good level; demonstrated a detailed knowledge of the associated subject matter; achieved a very good level of the skills required for the subject matter.

50% - 59%: The learner has: attained the module learning outcomes at a good level; demonstrated a good knowledge of the associated subject matter; achieved a good level of the skills required for the subject matter.

40% - 49%: The learner has: attained the module learning outcomes at a basic level; demonstrated a basic knowledge of the associated subject matter; achieved a basic level of the skills required for the subject matter.

<40%: The learner has not: attained the module learning outcomes; demonstrated sufficient knowledge of the associated subject matter; demonstrated a sufficient level of the skills required for the subject matter.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	36	
Practical	12	
Independent Learning	87	

Essential Material

- Muller, R. _Device Electronics for Integrated Circuits_. USA: Wiley, 2003.
- Pierret, R. _Semiconductor Device Fundamentals_. USA: Addison Wesley, 1996.
- Sze, S. _Physics of Semiconductor Devices_. USA: Wiley, 2006.

Supplementary Material

- "Institute of Electrical and Electronic Engineers." www.ieee.org

Requested Resources

- ENGINEERING LAB: Electronics